



Rome and the Medieval Period

WITH THE DEFEAT OF ATHENS BY SPARTA IN THE PELOPONNESIAN WAR (431–404 BCE), the Greek city-states began to disintegrate. By the time Aristotle died in 322 BCE, they had become part of the short-lived Macedonian Empire, founded by Alexander the Great. Republican Rome invaded shortly afterward, and they were eventually incorporated into the Roman Empire. The center of learning shifted from Athens to Alexandria in Egypt. In the uncertain years that followed, the confident theoretical speculations of the Greek naturalists and formalists became the object of Skepticism and Cynicism. The Hellenistic and later Roman period witnessed a turn to more practical philosophies of life, such as Epicureanism and Stoicism.

The Romans were great technologists and administrators, but contributed little to the development of science. Mystical forms of Neoplatonism became popular and informed the emergence of Christianity in the early days of the Roman Empire. When Christianity was accepted as the state religion, the works of pagan scholars such as Pythagoras and Aristotle were denigrated and condemned. With the decline of the Roman Empire, Western Europe entered what is known as the Dark Ages, a time when many of the classical Greek texts were destroyed or lost. Alexandrian scholars fled to Constantinople, then to Persia, where the classical texts were rediscovered by Islamic scholars and eventually by Christian scholars with the advent of the Crusades. During the middle and later medieval period, Aristotle's natural philosophy was integrated with Christian theology, effectively fossilizing his theories as church dogma.

Although science developed little during the medieval period, the medievals were not as hostile to it as is commonly supposed. They did not generally persecute practicing scientists and did not burn hundreds of thousands of neurotic and psychotic women whom they misdiagnosed as witches. What was distinctive about the medieval period was the general lack of interest in the empirical evaluation of scientific theories, including psychological theories. Most medieval scholars were content to develop their theories based upon classical and theological authorities.

THE ROMAN AGE

The Roman Age began with the 500 years of the Roman Republic prior to the time of Julius Caesar (100–44 BCE), during which the Senate governed Rome. After Caesar's assassination, Rome was governed by a series of emperors, beginning with Augustus (63 BCE–14 CE). The Roman Empire, which at the height of its power stretched from the British Isles to the Middle East, brought stability and order to the Mediterranean world for nearly 400 years, until it collapsed due to internal pressures and external invasions.

The Hellenistic Period

During the Hellenistic period (the time between the dissolution of the Greek city-states and the emergence of the Roman Empire), the theoretical speculations of early Greek thinkers were rejected. **Skeptics**, such as Pyrrho (c. 365–c. 275 BCE), repudiated all pretensions to knowledge. They advocated the suspension of belief and recommended that people follow the local moral and religious practices prevalent in society at any particular time and place. The **Cynics** Antisthenes (c. 445–c. 364 BCE) and Diogenes (c. 412–323 BCE) dismissed classical learning and conventional morality and recommended a life of natural independence, free of government, custom, and tradition. They were called Cynics by virtue of the extremely primitive form of life they advocated and exemplified. *Cynic* means “doglike,” and Diogenes is reputed to have behaved like a dog, using the public square as his toilet and venue for masturbation.

Hellenistic thinkers turned from theoretical speculation to more practical philosophies of life. These had a significant impact on Roman religious practices and moral attitudes, but contributed little to science and psychology. Epicurus (c. 341–270 BCE) and his Roman disciple Lucretius (c. 99–55 BCE) developed a philosophy of individual happiness grounded in Democritean atomism and the denial of the possibility of an afterlife. **Epicureanism** was not a simple hedonism of maximizing pleasure through wine, women, and song, but a philosophy of moderation based upon reason, choice, and discipline, which would supposedly ensure the greatest amount of genuine happiness over the long term. The highest form of pleasure was held to be friendship, and rich food and drink (especially in excess) were to be avoided. As for sex, according to Epicurus, it never did anyone any good, and you should count yourself lucky if you were not harmed by it!

The philosophy of Zeno of Citium (c. 333–264 BCE) was also based upon Democritian atomism and came to be known as **Stoicism** (because the school in which Zeno taught had a *stoa poikile*, or multicolored covered gallery). According to Zeno, everything in nature is predetermined according to a divine plan. Everyone is assigned a role and destiny prescribed by God, and virtue consists in acting in accord with

this natural law and adopting the right attitude toward it. The good life consists of freely accepting one's fate, whether it be good or ill, with indifference. Stoicism had widespread appeal among the Romans and seemed especially suited to their moral, social, and practical temperament. Emperors (Marcus Aurelius, 121–180 CE), statesmen (Seneca, c. 4 BCE–65 CE), and slaves (Epictetus, c. 55–c. 135 CE) embraced it.

Alexandrian Science

As theoretical speculation declined in Greece, the center of science and learning shifted from Athens to Alexandria, founded by Alexander the Great and completed by Ptolemy, one of his generals. The Ptolemaic dynasty created a great university and a famous library. The Great Library of Alexandria, which contained over a quarter of a million volumes, was one of the seven wonders of the ancient world.

Herophilus of Chalcedon (335–280 BCE), a Hippocratic disciple, founded the anatomical school at the Museum of Alexandria, where medical research and teaching were based upon the dissection of human cadavers and the vivisection of animals. Herophilus distinguished between the sensory and motor nerves (von Straden, 1989), and with his colleague Erasistratus (c. 304–c. 250 BCE) he explored the functions of the nervous system. By exposing and severing nerve bundles, they determined that the nerves are responsible for transmitting impulses (or “motions”) from the senses to the brain and from the brain to the muscles; they speculated that these impulses were borne by animal spirits (very small material particles) passing through otherwise hollow nerves. The Roman physician Galen later developed this speculation into a detailed theory of nervous transmission that remained influential until the 17th century. Herophilus is sometimes called the father of anatomy (and Erasistratus the father of physiology), but his pioneering work also brought dissection into disrepute. The Christian father Quintus Septimius Florens Tertullian (c. 160–230 CE) accused him of having dissected live criminals, supplied by King Ptolemy I, who reputedly also participated in many dissections (Roach, 2003). The Romans consequently prohibited the practice in Alexandria and throughout the empire.

Euclid (c. 325–c. 265 BCE) was curator and librarian of the Great Library of Alexandria. His *Elements* integrated and organized the body of arithmetic and geometric knowledge developed by the Egyptians, Babylonians, and Pythagoreans, within which he derived theorems such as Pythagoras's theorem from self-evident axioms. Euclid's deductive system of demonstration, in which the truth of theorems is shown to follow logically from the assumed truth of axioms, became a popular model for theoretical explanation in natural science. Isaac Newton (1642–1727) later employed it in his statement of the theory of universal gravitation (Newton, 1687/1969), as did Thomas Hobbes (1588–1679) in his mechanistic theory of human psychology and behavior (Hobbes, 1640/1966) and Clark L. Hull (1884–1952) in his neobehaviorist theory of learning (Hull, 1943).

Archimedes (c. 287–c. 212 BCE) founded the science of hydrostatics with his treatise *On Floating Bodies*. He was reputedly in the public baths when he invented a method for measuring the specific gravities of substances by displacement (in order to identify forgeries among the royal jewelry) and ran naked through the streets shouting “Eureka!” (“I have it!”). In *On Plane Equilibriums*, he organized the known principles of mechanical equilibrium into a deductive system, from which he derived theorems governing the (idealized) operation of mechanical levers. Archimedes put his theoretical knowledge to practical use in his design of military catapults, when he returned to his native Syracuse (in Sicily) to serve as a military engineer in the defense of the city as the Romans besieged it. According to legend, he was slain by Roman soldiers while he contemplated a geometric problem.

It was also in Alexandria that Claudius Ptolemy (c. 100–c. 170 CE) developed Aristotle’s geocentric (Earth-centered) theory of the heavens in his *Almagest*. He claimed that the planets maintain circular orbits around a fixed Earth, but introduced a system of epicycles (circles within circles) to account for the retrograde motion of some of the planets (which appear to slow down, stop, reverse, and then move on again in the original direction). Ptolemy dispensed with Aristotle’s crystalline planetary spheres, but retained his celestial sphere, which supposedly carried the fixed stars. This astronomical system was generally accepted (with only minor modifications) until the 16th century, when it was challenged by pioneers of the scientific revolution in Europe such as Nicolaus Copernicus (1473–1543), Johannes Kepler (1571–1630), and Galileo Galilei (1564–1642). Ptolemy also did pioneering experiments on color and the reflection and refraction of light, which are documented in his *Optics* (Smith, 1996).

Rome and Science

Although the Romans produced many technological marvels, such as aqueducts, baths, roads, and military machines, they created little in the way of developed natural or psychological science. The Romans were great engineers and mechanics, insofar as it suited them in the practical world of everyday affairs, but they seem to have been constitutionally uninterested in the pursuit of abstract thought and speculative theory. They were not actively hostile to science and respected and preserved the works of the classical Greeks. Roman scholars transcribed the works of Plato and Aristotle and translated them into Latin, and aristocratic Romans sent their children to study in Alexandria. However, with few exceptions, such as the philosopher Lucretius (c. 99–c. 55 BCE), who developed the atomism of Epicurus in his poem *De Rerum Natura*, they expressed little interest in expanding on the works of their predecessors. Science entered a period of progressive decline and was eventually suffocated by the development of Christianity, which became the official religion of the Roman Empire in 380 CE.

Galen Once again, the physicians were a notable exception. Galen (c. 130–c. 200 CE), who became personal physician to the Roman Emperor Marcus Aurelius, was trained in Alexandria, where medical science was originally based upon the dissection of human cadavers. Although by Galen’s time human dissection was prohibited throughout the Roman Empire, he managed to supplement his theoretical medical training with experience as a gladiatorial surgeon and with his own dissection of animals such as goats, pigs, sheep, and cattle.

Like Aristotle, Galen maintained that every structural form, organ, and system of the human body has its own distinctive purpose or function, which he detailed in *On the Usefulness of the Parts of the Body*. One especially influential doctrine was Galen’s account of the role of “vital” and “animal” spirits in supporting biological and psychological functions. According to Galen, the heart is responsible for the distillation of vital spirits drawn from the air, which regulate movement. When conveyed to the network of interwoven blood vessels in the brain known as the *rete mirabile* (the “marvelous net”), they are further refined to animal spirits, which are responsible for perceptual and cognitive functions. Galen claimed that the animal spirits fill the ventricles (cavities) of the brain and the nerves and that they issue from the eyes in vision.

The principles of Galen’s anatomy and physiology dominated medical science in the ensuing centuries and became virtual dogmas during the medieval period. They persisted well into the 17th century: Theories referencing animal spirits are to be found in the mechanistic physiology of Hobbes and Descartes, for example. One particularly influential doctrine was Galen’s development of Hippocrates’ theory of the four bodily humors into a theory of personality types (Figure 3.1).

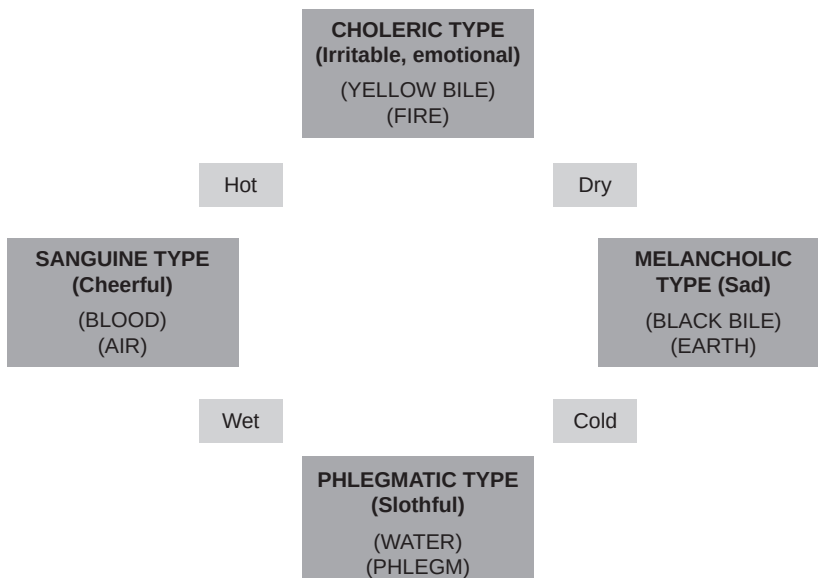


FIGURE 3.1 Galen: Personality types.

According to this theory, the cheerful, or **sanguine**, type of personality, has an excess of blood; the sad, or **melancholic**, type has an excess of black bile; the irritable and emotional, or **choleric**, type has an excess of yellow bile; and the slothful, or **phlegmatic**, type has an excess of phlegm. This theory has long since entered the vernacular and survives in everyday phrases such as “phlegmatic character” and “bad humor.” In *On the Diagnosis and Cure of the Soul's Passions*, Galen recommended individual counseling for emotional problems and documented the physiological symptoms of the “love sickness,” such as increased pulse and heart rate.

Neoplatonism

The Pythagorean elements of Plato's philosophy were also developed in the early years of the Roman Empire. **Neoplatonic** theories focused on the mystical and spiritual elements of Plato's philosophy rather than its critical rationalism and exerted a powerful influence on the early development of Christianity.

Philo of Alexandria (c. 20 BCE–50 CE) claimed that knowledge is revealed by God and cannot be attained through sense experience or reason. He maintained that both sense experience and reason are impediments to knowledge, which can be attained only via the passive reception of divine illumination by a purified mind, through meditation, trance, or dreams. Like Plato, Philo claimed that the immaterial soul is “imprisoned” in the inferior material body. Throughout a human life it can either rise above carnal experience and move toward the light (of divine illumination) or sink down to carnal experience and move away from the light.

The most influential Neoplatonist was Plotinus (c. 204–270 CE). Educated in Alexandria, he founded a philosophical school in Rome under the protection of the Emperor Gallienus (c. 213–268 CE). He also conceived of the immaterial psyche as imprisoned in the inferior material body, from which it could escape only through transcendental experiences, such as meditation and dreams. In the *Enneads*, Plotinus claimed that all reality is based upon a series of emanations from “the One,” the original and eternal source of being. Next comes the realm of intelligence (a reflection of the One or ideas in the mind of the One), psyche, and finally matter. Like Plato, Plotinus claimed that the material world is an inferior copy of the divine abstract realm, although he was less critical of sense experience than Philo and Plato. He suggested that sense experience could be a source of beauty in art and music, as well as a source of ignorance and evil. He claimed that sense experience provides an accurate representation of external physical particulars, but since they are constantly changing, it is of little value. Genuine knowledge can be attained only through the apprehension of ideas in the realm of intelligence or, in rare cases, through mystical union with the One (O'Meara, 1995).

Like Philo, Plotinus maintained that the soul can ascend to spirituality and mystical unity or descend to the carnal degradations of the material body. He was so committed to the view that the soul is imprisoned in the material body that he took no care of his own body. He ate little, abstained from sex, and was indifferent to matters of personal hygiene. He left Rome after the assassination of Gallienus and died of leprosy shortly afterward.

The Decline of the Roman Empire

The Roman Empire lasted from approximately the first to the fifth century CE, although it was in serious decline by about the second century. Scientific thinking degenerated, even in Alexandria, after the Romans took over the administration of Egypt following the defeat of Cleopatra and Mark Anthony at Actium in 31 BCE (and their subsequent suicides). The government corruption, economic degeneration, and barbarian invasions that marked the decline of the Roman Empire led many to seek solace in “other-world” philosophies, such as Neoplatonism and Christianity.

Historians mark the end of the Roman Empire with the deposition of the Emperor Romulus Augustus in 476 CE. A remnant continued in the East, founded by the Emperor Constantine (c. 272–337 CE), with its center at Byzantium, which was later renamed Constantinople. The Byzantine Empire lasted until 1453 CE, when Constantinople fell to the Turks. It contributed little to the development of science, but for a period preserved much of the classical scholarship that was destroyed or lost in the West.

Christianity The disintegration of the Roman Empire was paralleled by the development of “mystery” religions, which promised otherworldly salvation in times of great trouble and tribulation. The one that came to dominate was based upon the life and teachings of Jesus (c. 4 BCE–30 CE). Early doctrinal dissension, often extremely violent, was settled through the early church councils, such as the Council of Nicaea in 325 CE. Originally persecuted, Christians gained increasing power and influence after the Emperor Constantine’s Edict of Milan in 313 CE, which granted religious toleration. The Emperor Theodosius (c. 346–395 CE) proclaimed Christianity as the official state religion in 380 CE and prohibited all pagan religions.

All this was extremely bad news for learning in general and scientific thought in particular. Pagan science was condemned along with pagan religion. The Archbishop Theophilus of Alexandria, who like many early Church fathers was zealous in his destruction of all pagan symbols, was responsible for the destruction of much of the Great Library at Alexandria in 391 CE. Hypatia (c. 370–415 CE), the distinguished female mathematician and astronomer, was murdered by a mob of

Christians. They dragged her into a church and ripped the flesh from her body with roof tiles, most likely on the orders of Cyril, the nephew of Theophilus, who had replaced him as patriarch of Alexandria. Cyril was later canonized. Many scholars fled Alexandria, which went into decline as a center of learning and science.

Some sought refuge in Athens, where Plato's Academy was still in operation, although by this time devoted almost exclusively to mystical speculation. However, Christian pressure persuaded the Emperor Justinian (483–565 CE) to close it down in 529 CE and to forbid the study of all "heathen learning." Other scholars migrated to Constantinople, taking many classical works with them. Despite Constantine's noble intention to create a virtuous Christian city in the East, the debauchery of the masses and the ferocity with which every new type of heresy was persecuted forced many Alexandrians to flee to Mesopotamia and eventually to Persia. There they translated the works of Plato, Aristotle, Euclid, Archimedes, and Ptolemy, making the works of these classical theorists available to the Islamic Empire that was soon to engulf them.

Christianity and Pagan Thought One of the earliest debates within Christianity was whether to dismiss alternative philosophies and religions as pagan and heretical, the position championed by Saint Jerome (c. 345–420 CE), or whether to integrate at least some elements of them within Christianity, the position championed by Saint Ambrose (c. 340–397 CE). Although Jerome represented the original consensus, Ambrose and his followers eventually won the day. As a result, many Christians enjoy such originally pagan symbols, artifacts, and offices as holly wreaths, Christmas trees, incense, and the originally Druid roles of best man (strictly, next best man, since his duty was to take the place of the groom if the groom died or reneged) and maid of honor (whose virginity was sacrificed to the next best man if the groom did his duty). One significant consequence was that the bishop of Rome adopted the pagan office of Pontifex Maximus, a position that evolved into the authoritarian structures of the papacy. However, these early Church fathers did not look to Aristotle's empirical science and materialist psychology, but embraced the Neoplatonic conception of the psyche as an immortal spiritual entity temporarily imprisoned in an inferior material body.

Saint Augustine Augustine of Hippo (354–430 CE) was largely responsible for the Neoplatonization of Christianity. He affirmed the conception of the soul as a separate spiritual entity temporarily imprisoned in an inferior material body. He reaffirmed Plotinus's view that knowledge can be attained only through acquaintance with the eternal forms or ideas and the illumination of God and that man should turn away from the world of the senses and carnal pleasure.

Born in North Africa to a pagan father and Christian mother, Augustine converted to Christianity at the age of 31. After years of debauchery, during which he "boiled over in . . . fornications," he experienced a revelation and dedicated the

rest of his life to monastic devotion. He is often remembered for his prayer from the licentious days prior to his revelation in which he pleaded: “Give me chastity and continency, only not yet.” Augustine became bishop of Hippo in 396 CE and advanced a number of doctrines that were to dominate Christian theology into the 12th century, such as the doctrines of free will, the reality of the fall, and original sin.

Augustine developed a form of **substance dualism** and a set of supporting arguments that later came to be associated with René Descartes (whom he almost certainly influenced). He maintained that the soul is a special and simple spiritual substance, which is distinct from material substance and can survive bodily death. Our knowledge of the distinctive nature of the soul as a special type of “thinking substance” is based upon the certainty of our beliefs about ourselves, in contrast to our often erroneous beliefs about physical objects in the external world. Augustine argued that he could not be mistaken about his existence, for he must exist in order to be mistaken:

The certainty that I exist, that I know it, and that I am glad of it, is independent of any imaginary and deceptive fantasies.

In respect of these truths I have no fear of the arguments of the Academics. They say, “Suppose you are mistaken?” I reply, “If I am mistaken, I exist.” A non-existent being cannot be mistaken; therefore I must exist, if I am mistaken. Then since my being mistaken proves that I exist, how can I be mistaken in thinking that I exist, seeing that my mistake establishes my existence?

—(*The City of God*, pp. 459–460)

Augustine extolled the potential of the “inner senses” to provide us with self-knowledge of the contents of experience, thought, and memory and maintained that self-knowledge of the soul leads to knowledge of God. He created a new literary art form in his biographical *Confessions*, which provided an exhaustive catalogue of his sins with an extensive psychological commentary upon them. It contains astute observations on emotion, the perception of time, memory, and dreams. Augustine maintained that certain forms of knowledge are innate, such as our knowledge of mathematical relations and moral principles, and noted that certain forms of memory, such as our memory of emotions, do not involve images.

Like Philo, Augustine depreciated reason and experience in the pursuit of knowledge. He maintained that they are of value only insofar as they accord with Christian theology:

For whatever a man has learned elsewhere is censured there [Holy Scripture] if it is harmful; if it is useful, it is found there.

—(*On Christian Doctrine*, p. 63)

Augustine advocated and exemplified a life of piety and humility. He championed faith and emotional communication with God over reason and sense experience and recommended that humankind turn away from the ways of the world and look to God and the promise of Heaven.

The Fall of the Roman Empire

It is not surprising that Augustine held these views or that many accepted his theological vision in the ensuing centuries. The Western Empire was in disarray, ravaged by war, famine, and disease. In 410 Rome was sacked by the Visigoths under Alaric and later by the Huns and the Vandals. Communication and the rule of law broke down in the West. Augustine wrote *The City of God* (from 413 to 427) in response to Alaric's sacking of Rome, contrasting the temporal earthly city dominated by materialism and evil with the eternal and spiritual city of God that embodies goodness and salvation.

The new tribes that settled the fragments of the Western Empire, such as the Vandals, the Angles, the Saxons, and the Jutes, could not maintain the commercial centers of the former empire, and western Europe became increasingly rural and feudal. City walls eroded, harbors silted, and roads fell to ruin. Treasuries were looted, and learned books were destroyed or lost. The social unit was the farm-estate, which sustained a subsistence agricultural economy. Trade diminished to a vanishing point, and populations declined. Rome was reduced from a population of 1,500,000 in the first century to 300,000 in the fifth century, and by the end of the sixth century only about 50,000 remained in the ruins and rubble. The money economy returned to a barter economy.

MEDIEVAL PSYCHOLOGY

It is customary to mark the beginning of the **medieval period** with the deposition of Romulus Augustus in 476 and its end with the fall of Constantinople to the Turks in 1453. There is, of course, a high degree of arbitrariness about these dates. The Roman Empire was in decline long before the deposition of Romulus Augustus, and there were anticipations of the return of scientific thought in the centuries before the fall of Constantinople. Still, these dates are not without foundation. By the end of the fifth century the money-based commercial empire of the Romans had dissolved into isolated feudal enclaves based upon a rural subsistence economy. Most of the classical works were destroyed or lost and did not become available to Western scholars until the 12th century. By the end of the 15th century the Renaissance and Reformation were in full swing in western Europe, and the antipathy to tradition that they represented was a harbinger for the scientific revolution that began in the 16th century.

In western Europe, the period from about 500 to 1000, during which commerce and learning declined, is usually characterized as the early medieval period (the term *medieval* means “middle ages”). The period between about 1000 and 1300, during which commerce and learning revived, is usually characterized as the middle medieval period. The period from about 1300 to 1600, during which the intellectual and social system of the medieval world broke down through internal inconsistencies and the pressure of the Renaissance and Reformation, is usually characterized as the late medieval period.

It is customary to represent the medieval Christian Church as hostile to scientific thinking, to the point of the active persecution of scientists. Although there is some truth in this picture, it is also somewhat of a caricature, and the real story is more complex and interesting. Moreover, whatever learning was preserved in the West, particularly during the early medieval period, sometimes known as the Dark Ages, was preserved by clerics in monastic enclaves. However, during this period civilization and learning stagnated in Europe, apart from a brief renaissance during the reign of Charlemagne (742–814).

Islam

It was the Islamic Empire that rediscovered, translated, and preserved the works of Aristotle and other Greek thinkers. The prophet Muhammad was born in 570 and experienced his first vision of the archangel Gabriel in 610. It was revealed to him that he had been chosen as the messenger of Allah (God), and he received the sacred writings that formed the basis of the Koran, the holy book of the new Islamic religion. His followers quickly captured the holy cities of Mecca and Medina. Spectacularly, within a hundred years of the death of the prophet in 632, the Islamic Empire extended over an area greater than the Roman Empire at its height, including North Africa and Spain, Syria, Egypt, Arabia, and Persia. This brought Islamic scholars into contact with the classical works lost to the Western world.

The Islamic conquerors initially condemned and rejected such pagan works with the zeal of the early Christians. They destroyed the remains of the Great Library at Alexandria in 642, using the classical volumes to fuel the city baths. In justification, Caliph Omar claimed that “if these writings of the Greeks agree with the book of God, they are useless, and need not be preserved; if they disagree, they are pernicious, and ought to be destroyed,” an attitude almost identical to that of Augustine.

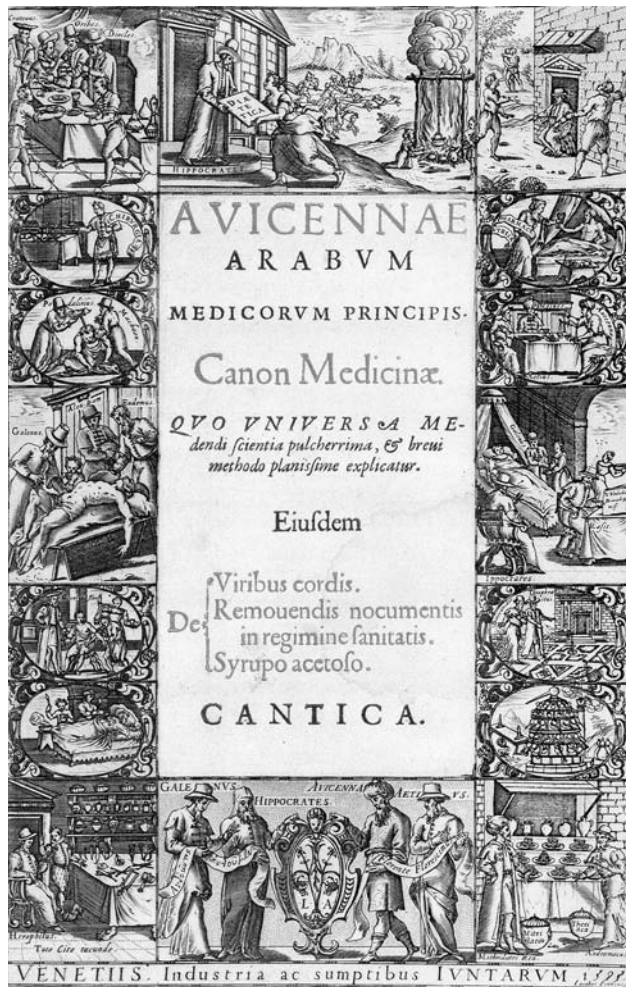
Later caliphs were less antagonistic. Around 800 Caliph Haroun-al-Raschid had the works of Aristotle, Hippocrates, and Galen translated into Arabic; and his successors sent missions to Constantinople and India to discover other scientific works suitable for translation. The works of Plato, Plotinus, Euclid, Archimedes,

and other classical writers became available to the Islamic Empire when it conquered Persia (where the works had been preserved by Alexandrians who had fled to Persia from Constantinople), and scholars such as Yaqub ibn Ishaq al-Kindi (d. after 866) and Abu Nasr al-Farabi (c. 870–950) translated them into Arabic. In this fashion the Islamic Empire became the repository of classical learning and science. The most famous Islamic scholars were Abu Ali al-Husayn ibn Sina (c. 980–1037), known in the West as Avicenna, and Muhammad ibn Roshd (1126–1198), known in the West as Averroës, who produced translations of, and commentaries on, these classical works. Avicenna and Averroës tried to integrate the central features

of Platonic and Aristotelian philosophy with Islamic theology. Abu Ali Hasan ibn al-Haitham (965–1039), known in the West as Alhazan, developed the first detailed and experimentally grounded theory of visual perception.

Avicenna Avicenna was a physician to several Persian princes. He developed Galen's theories in his encyclopedic *Canon of Medicine*, a text that was widely used in medical schools in Europe and Asia during the medieval period. He wrote extensive and influential commentaries on the works of Aristotle, including psychological treatises based upon Aristotle's *De Anima* (Gutas, 1988). He became known in the Islamic world as the “third Aristotle,”¹ although his general Aristotelian principles were heavily larded with Platonic modifications (Goodman, 1992).

Avicenna developed the hierarchical account of the nutritive, sensitive, and rational psyche to be found in Aristotle's *De Anima*. He also followed Aristotle in distinguishing between passive and active reason and maintaining that active reason is immortal. However, in contrast to Aristotle, who claimed that passive reason is materially instantiated,



Avicenna: *Canon of Medicine*, Venice 1595.

¹Abu Nasr al-Farabi, who introduced Aristotelian logic to Islam, was known as the “second Aristotle.”

Avicenna was a dualist who claimed that passive reason is a capacity of the immaterial or spiritual psyche. This conception was famously developed in Avicenna's **flying man argument**, which, like Augustine's similar argument, anticipated the argument for substance dualism later developed by René Descartes. Avicenna claimed that if a full-grown person suddenly came into existence, suspended in space, with limbs separated and eyes covered, he would have no sensation but would nevertheless be aware of his existence as an entity distinct from his body:

Suppose that he was just created at a stroke, fully developed and perfectly formed but with his vision shrouded from perceiving all external objects—created floating in the air or in space, not buffeted by any perceptible current of the air that supports him, his limbs separated and kept out of contact with one another, so that they do not feel each other. Then let the subject consider whether he would affirm the existence of his self. There is no doubt that he would affirm his own existence, although not affirming the reality of any of his limbs or inner organs, his bowels, or heart or brain, or any external thing. Indeed he would affirm the existence of this self of his while not affirming that it had any length, breadth or depth. And if it were possible for him in such a state to imagine a hand or any other organ, he would not imagine it to be part of himself or a condition of his existence.

—(Rahman, 1958, p. 16)

Avicenna followed Plato in treating forms or universals as prior to individual substances and identified Aristotle's prime mover with the Islamic God, or Allah. He also adopted a variant of Plotinus's theory of emanations to describe the relation between God, the world of intelligible ideas, immaterial souls, and material bodies.

Averroës Averroës was a judge and physician who spent most of his life in Spain. He wrote detailed commentaries on Aristotle, whom he greatly venerated, and became known as "The Commentator" (on Aristotle, who had become known as "The Philosopher"). He provided an interpretation of Aristotle that was generally devoid of the Platonic modifications made by Avicenna (Kogan, 1985). He followed Aristotle in treating individual substances as prior to forms or universals, and his interpretation became foundational for many medieval Christian apologists, save for one problematic aspect. Averroës treated active reason as immaterial and immortal, but did not equate it with the individual human psyche. He argued that there is no way to distinguish the active reason of different human beings if active reason has no physical properties or spatial location and consequently maintained that active reason in all humans is numerically identical: It is one and the same in all humans. This doctrine, "that the intellect of all men is one and the same in number," became known in the Christian world as the **Averroës heresy** and was

condemned by the Bishop of Paris in 1270 (Thorndike, 1944). Since he followed Aristotle in treating the rational psyche of human beings as materially instantiated, Averroës maintained that he could not accept the immortality of the soul on rational grounds, although he embraced it as an act of faith (Leaman, 1988).

Alhazan Alhazen was perhaps the first to maintain that vision occurs when light reflected from external objects enters the eye. He conducted original experiments on light reflection and refraction and atmospheric effects on vision. His *Book of Optics* (1021) contains detailed discussions of color perception, apparent size, and double vision. Alhazen explored the problem of image inversion and located binocular vision in the “common nerve” (the optic nerve). He distinguished between sensation and perception and was one of the first theorists to relate the physics of light reflection to the anatomy of the eye. His empirical work and the problems it raised set the medieval agenda of research in visual perception (Lindberg, 1968).

European Recovery: Reason and Faith

The so-called Dark Ages in western Europe came to an end around 1000. Economies began to improve; the population began to increase again, particularly in the cities; the feudal system and the papacy evolved into dominant and integrative social structures. Commerce and communication improved, and theory and learning experienced a revival of sorts. Teachers of law, grammar, rhetoric, and logic were in high demand in urban centers, as religious and civic authorities tested the limits of their jurisdictions. Students and masters formed themselves into corporate entities called universities, with defined powers and rights. These were founded at Bologna (1088), Paris (1150), Oxford (1167), Padua (1222), Salamanca (1218), Vienna (1365), Prague (1348), and other urban centers, often in conjunction with the expansion in cathedral construction. Toward the end of the medieval period universities began issuing certificates, such as the baccalaureate, master’s, and doctoral degrees (Pyenson & Sheets-Pyenson, 1999).

The first Crusade (1095) brought the Western world in contact with Islamic scholarship, which had preserved the works of early Greek thinkers. The medical schools of many medieval universities adopted Avicenna’s *Canon of Medicine* as their primary text, and the commentaries of Avicenna and Averroës introduced the medieval Christian world to the works of Aristotle. Although condemned in 1220 and 1277, the naturalism of Aristotle eventually came to displace the Neoplatonism of Augustine as the conceptual foundation of Christianity.

The Church had by this time become more open to forms of knowledge other than scripture and revelation. Saint Anselm (1033–1109) claimed that reason and sense experience can supplement faith and developed famous arguments

purporting to demonstrate the existence of God. Peter Lombard (c. 1095–1160) claimed that knowledge of God can be obtained through knowledge of divine works, including the natural world and its human inhabitants. Saint Albertus Magnus (c. 1193–1280) produced a comprehensive review of the works of Aristotle and his Islamic commentators and recommended reason and sense experience as legitimate sources of knowledge, since he presumed that neither would conflict with scripture.

Peter Abelard Peter Abelard (1079–1142) also produced translations and reviews of Aristotle's works. He raised the employment of reason and argument to new heights and is often credited with the revival of the dialectic method of the early Greek philosophers, notably in his *Sic et Non* ("For and Against"). Abelard was convinced that any method of argument, including critical dialectic, would affirm God's existence, goodness, and wisdom. However, in practice his arguments exposed a number of conflicting theological positions, which got him into trouble with the church authorities.

While a canon of Notre Dame Cathedral in Paris, he met Héloïse (1101–1164). By his own account, Abelard immediately set out to seduce her. He could scarcely believe his own good fortune when Héloïse's uncle, Fulbert, offered free room and board in his house if Abelard would agree to tutor his niece privately. According to Abelard, he might as well have "entrusted a tender lamb to a hungry wolf" (Robertson, 1972, p. 43). He seduced her, and for the next few months the lovers "left no phase of love untried" (Grane, 1970, p. 49). Eventually Fulbert figured out what was going on and threw Abelard out, although by this time Héloïse was pregnant.

The lovers married in secret. When Fulbert publicized the marriage, Abelard made Héloïse pretend to accept holy vows. He took her to a convent and dressed her in a nun's habit. Fulbert was infuriated by Abelard's shoddy attempt to hide his own transgression, and with some aides, attacked Abelard in bed one night and castrated him: "They cut off those parts of my body with which I had done the deed they deplored" (Robertson, 1972, p. 55).

The passion of the lovers continued unabated, although now on a rather more Platonic level, as testified by the love letters they exchanged in the following years. Héloïse became abbess and prioress of the Paraclete Abbey, which Abelard founded. Héloïse's letters to Abelard include the learned *Problemata Héloissae*, which took the form of philosophical questions (with some spicy interludes). Abelard continued to court controversy and attention. A church council condemned his work in 1121, and he barely survived an assassination attempt in 1132. He was censured by Pope Innocent II in 1140 and ordered to cease writing and teaching. He died two years later and was buried in the Paraclete Abbey; on her death, Héloïse was buried beside him. Their remains were later transferred to a crypt in the Père-Lachaise cemetery in Paris.

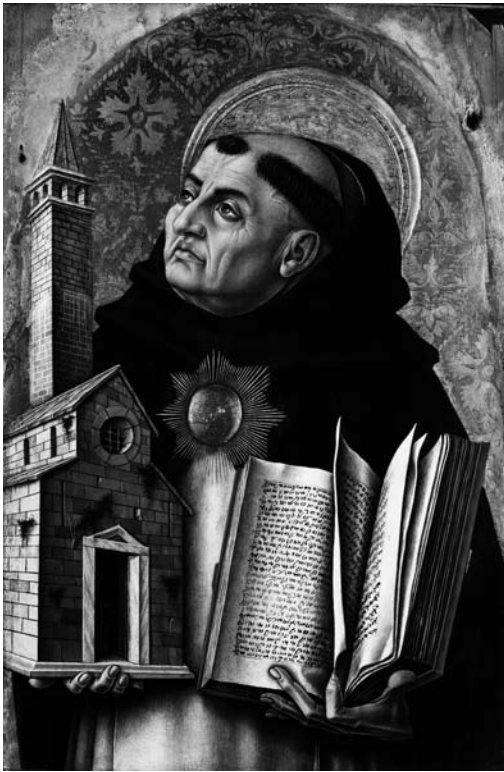
The Christian Church and Aristotelian Philosophy

The Christian Church originally rejected the works of Aristotle, which were condemned by Saint Bonaventure (1221–1274). However, medieval **scholastics** quickly followed their Islamic counterparts by trying to integrate them with Christian theology. In the prevalent literary metaphor of the time, they tried to effect a kind of intellectual “marriage” of Aristotelian theory and Christian theology. Aristotle’s active reason was identified with the immortal soul, and Aristotle’s unmoved mover was identified with the Christian God.

Thomas Aquinas The classic statement of this attempt to reconcile reason (or at least the reason of Aristotle) and faith is to be found in the work of Saint Thomas Aquinas (c. 1225–1274). Born in the Italian city of Roccasecca, he was known as the “dumb ox” to his fellow students and the “angelic doctor” to his later admirers. His original attempt to enter the Church was thwarted by his family, who imprisoned him in the hope of persuading him to change his mind. When all attempts (including sexual

temptation) failed, they relented and allowed him to travel to Cologne and study with Albertus Magnus, who persuaded him of the virtues of Aristotelian philosophy. Aquinas attained his master’s degree from the University of Paris, where he taught for a number of years. His major work, *Summa Theologica*, appropriated those elements of Aristotelian theory most congenial to Christian theology.

It is not hard to understand why Aristotelian theory appealed to Christian theologians. Aristotle’s universe was teleological and hierarchical. Every inanimate and animate object had its own end or function, and it was easy to interpret this in terms of divine purpose and creation. Aristotle’s *scala naturae* was represented as a “great chain of being” (Lovejoy, 1936), beginning with inanimate objects and reaching up through the vegetative, animal, and human kingdoms to the angels and God above, with the earth at the physical center of the universe and man at its spiritual center. This fixed hierarchical conception of the natural order served conveniently to sanction the established social hierarchy of the Church and the feudal system, with the pope and kings at its apex and the laboring peasants at its base, everyone serving their proper and fixed purpose in the feudal structure of nobles, lords, and



Aquinas: the “marriage” of Christian theology and Aristotelian philosophy.

vassals. Although initially controversial, Aquinas's Aristotelian version of Christianity came to supplant the earlier Neoplatonic version promoted by Augustine and remains to this day the foundational theology of the Roman Catholic Church.

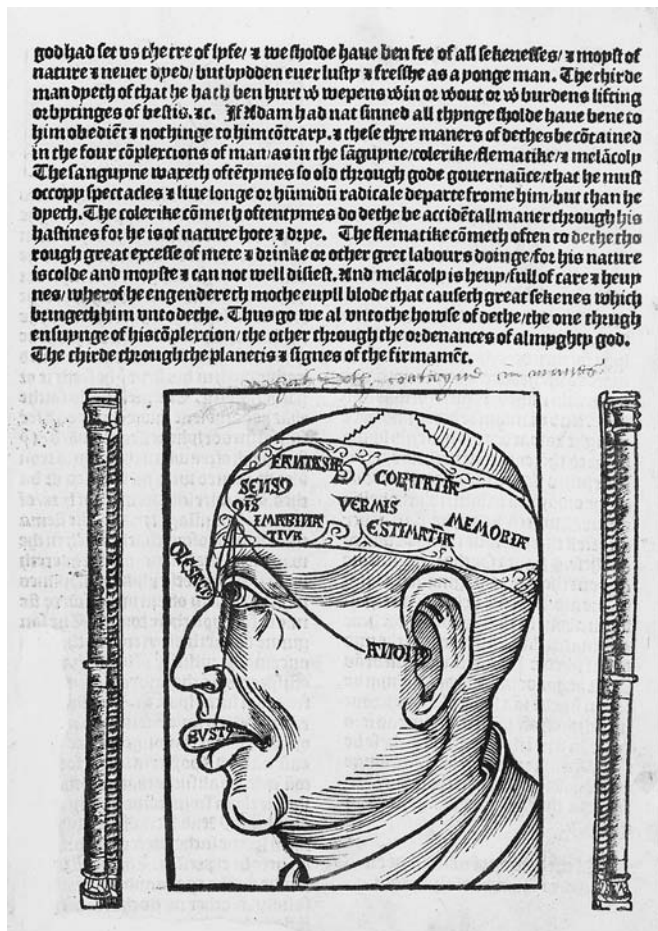
Aquinas revived the Aristotelian conception of the human psyche as the functional capacities of the human material body. He also went one step further and treated active reason as a functional capacity of the embodied human psyche (Abel, 1995; Haldane, 1992). Consequently, later Christian theology shaped by Aquinas has tended to focus on the sure and certain hope of resurrection rather than a spiritual afterlife. Like Aristotle, Aquinas claimed that thought is dependent upon sensory experience, since it is dependent upon the ability to form sensory images, and he consequently denied the existence of innate ideas (Davies, 1992).

Aquinas also recognized the **intentional** nature of psychological states such as thoughts, emotions, motives, and memories: The fact that such psychological states make reference to some object beyond themselves. Thus my thought that Aristotle was the first psychological scientist is directed to or about Aristotle. In the late 19th century, the German psychologist Franz Brentano (1838–1917) characterized intentionality as the distinctive “mark of the mental” (Brentano, 1874/1995).

The Inner Senses The early Christian fathers, such as Augustine, had followed Plato, Philo, and Plotinus in treating the soul as an immaterial spiritual entity temporarily imprisoned in the material human body, the view also embraced by Avicenna. Aquinas returned Christianity to the Aristotelian conception of the soul as the functional form of the material human body, the view embraced by Averroës, although both Christian and Islamic scholars in the medieval period tried valiantly to reconcile the two positions (Kemp, 1990). Given this significant change, neither Christian nor Islamic theological orthodoxy can be blamed for the substance dualism that dominated early psychological science. Post-medieval commitments to dualism were in a very real sense a by-product of the scientific revolution in Europe that began in the 16th century: This was the position championed by René Descartes, one of the primary advocates of the new mechanistic science.

In any case, although the differences between such theorists were theologically significant, they seem to have played a relatively minor role in the development of medieval psychological theories. Since so little was known about human physiology and neurophysiology, different views about the immaterial versus the material basis of the human psyche had little impact upon psychological theory, which may be one reason why few scientists were persecuted by religious authorities for their specifically psychological theories.

A good example is the popularity of the theory of the **inner senses** or “inner wits” throughout the medieval period (Harvey, 1975). This theory was an amalgamation of the psychology of Aristotle and the neurophysiology of Galen. The inner senses were usually identified as common sense, imagination, estimation, memory, and reason. They were held to be located in the **ventricles** (fluid-filled



Medieval depiction of inner senses. Inner Wits: The Nobell Lyfe and Natures of Man (c. 1521).

cavities) of the brain, which were identified with reasonable accuracy by Galen (although Galen himself believed that psychological capacities were instantiated in the substance of the brain rather than the ventricles).

Early versions of the theory are to be found in Nemesis (a fourth-century Christian physician) and Augustine (Green, 2003). Avicenna developed the most popular and influential version in his *Canon of Medicine*, the standard medical text of the medieval period (Kemp, 1997). Avicenna claimed that the three ventricles of the brain perform five distinct cognitive operations. The anterior ventricle receives impressions from the various sensory organs and nerves, which are integrated by the common sense located at the front of the ventricle; the images produced are stored by the imagination at the rear. The middle ventricle is responsible for both

the reconstruction of stored images to form complex representations (including representations of hitherto unobserved objects, such as men with wings or golden mountains) and estimation, based upon instinct or associative learning:

Then there is the estimative faculty located at the far end of the middle ventricle of the brain, which perceives the non-sensible intentions that exist in the individual sensible objects, like the faculty that judges that the wolf is to be avoided and the child to be loved.

—(Rahman, 1952, p. 31)

The posterior ventricle is responsible for memory of cognitive reconstructions and estimations produced in the middle ventricle.

The theory was employed to explain a variety of psychological phenomena beyond basic perception and cognition. The bizarre nature of dream images was explained in terms of imagination and memory operating independently of sense perception; and mania, melancholia, and accidie (a debilitating form of apathy) were attributed to disturbances of the different ventricles (Kemp, 1990). The theory was abandoned when the 16th-century anatomist Andreas Vesalius (1514–1564) demonstrated that the sensory nerves are connected to the rear of the brain and not to the anterior ventricle.

Medieval Christianity and Science

By the 13th century, Aristotle's works had been thoroughly assimilated by Christian theologians. This proved to be a mixed blessing. While Aristotle's achievements were duly recognized, the critical and empirical spirit behind them was not. Aristotle's generally cautious and qualified theories were elevated and fossilized into Christian dogma, to the point that it became a heresy to question those Aristotelian theories that were adopted by the Church, as Galileo, Bruno, and Descartes later learned to their cost.

However, it is a myth that the Church was actively hostile to science and used the Inquisition to stifle and inhibit those of a scientific and empirical bent. Few scientists are recorded as having being burnt at the stake by the Inquisition. The general response of the Church to problematic intellectual positions was the expurgation of printed works and excommunication of their authors (Thorndike, 1944). The Church was actively hostile to astrology, which was frequently practiced in conjunction with astronomy, but only one astrologer is recorded as having been condemned to death, one Cecco d'Ascoli in 1327 (Wedel, 1968).

It is also a myth that the medieval Church impeded the scientific study of medicine by prohibiting the dissection of human cadavers (Demaitre, 1975). The Council of Tours issued a prohibition against human dissection in 1163, but this was not directed toward medical science. It was introduced to discourage the

convenient practice of dismembering the mortal remains of Crusaders before they were shipped home. In any case, many medical schools, such as the Hippocraticum Medicorum Collegium at Salerno, the leading medical school in the 12th century, simply ignored the prohibition, and there was no systematic ecclesiastical opposition to human dissection during the medieval period. At many universities, such as the University of Bologna, human dissection was mandatory (Bullough, 1958). The students resisted it, not the university or church authorities (Kemp, 1990).

Guglielmo da Saliceto (1210–1277), an Italian surgeon, published a record of his dissections in 1275, as did Mondino de' Luzzi (c. 1275–1326) in 1316. One of the few anatomists to be sentenced to death by the Inquisition was Vesalius, who pioneered the scientific revolution in medicine at the University of Padua in the 16th century, at the very end of the medieval period. However, Vesalius was accused of murder before the Inquisition by the parents of a man he had supposedly dissected while still alive, not for his medical dissection per se. His life was spared by the intervention of Phillip II, and his sentence was commuted to a religious pilgrimage.

Pope Gregory IX instituted the **Inquisition** in 1233 to repress the remains of the **Catharian heresy** in Spain and the south of France, which Pope Innocent II had earlier launched the Albigensian Crusade to combat. The Catharists were critical of the wealth and power hierarchy of the Church and maintained that since Christ was poor the Church should follow his example and abandon its wealth. Many perished for that dangerous belief, but not for advocating any controversial scientific theory.

Witches and Demons Another myth is that the Church, largely through the offices of the Inquisition, condemned hundreds of thousands of unfortunate persons, mainly women, to burn as witches. The medieval period is often represented as a regressive period of reversion to superstitious theories of spirit possession and repressive treatments of psychological disorder. It is commonly supposed that during the medieval period symptoms of psychological disorder were treated as evidence of witchcraft or demon possession and that many innocents perished as a result of such ignorance (Altrocchi, 1980; Alexander & Selesnick, 1966; Suinn, 1975). Estimates of the human cost of this persecution are usually in the order of hundreds of thousands in Europe: “literally hundreds of thousands of women and children were condemned as witches . . . and burned at the stake” (Zax & Cohen, 1976, p. 41).

Yet, for most of the medieval period, the Church did not recognize the existence of witches and reserved the stake for unrepentant heretics (Kirsch, 1978). Although the Office of the Inquisition was undoubtedly repressive and employed secret investigations and torture, it was only employed to investigate witchcraft toward the end of the medieval period (Cohen, 1975). The famous tract against witchcraft, *Malleus Maleficarum* (*The Witches Hammer*), written by Jacob Sprenger



Burning witches.

and Heinrich Kramer, was published in 1487, years after the fall of Constantinople and the beginning of the European Renaissance, which are usually held to mark the end of the medieval period (Cohen, 1975).

Although this “huntsman’s bible” is often held to have been responsible for “hundreds and thousands of women and children being burned at the stake” (Alexander & Selesnick, 1966), the numbers are greatly exaggerated (Schoeneman, 1977), possibly by a factor as high as 100 percent (Maher & Maher, 1985; Trevor-Roper, 1967). For example, the extremely active Grand Inquisitor Bernard Gui (c. 1307–1323) dealt with 930 cases in his lifetime, of which 80 accused were already deceased when they came to trial: in only 43 cases were the accused condemned to the stake (Coulton, 1961, cited in Kemp, 1990). It has been argued that the European witch craze was not a function of medieval superstition and ignorance, but a product of the scientific revolution of the 16th century, which encouraged the idea that there might be empirical indices of demon possession and witchcraft (Kirsch, 1978).

The European “witch craze” was real enough, but reached its zenith in the 16th, 17th, and 18th centuries. Critics such as Cornelius Agrippa (1486–1535) and Philippus Paracelsus (1493–1541) challenged explanations of abnormal behavior in terms of witchcraft from the moment they were embraced by later Protestant and Catholic zealots. Johann Weyer (1515–1588) in the *Deception of Demons* (1563) and Reginald Scott (c. 1538–1599) in the *Discovery of Witchcraft* (1584/1971) were among the first to suggest that some of the persons identified as witches might be suffering from some form of psychological disorder.

Still, it is doubtful if all those who were burned as witches were psychologically abnormal, since the motives of their persecutors appear to have been many and various. They included social, political, economic, legal, and personal as well as religious reasons (Schoeneman, 1977; Spanos, 1978). For example, while most of the witches burned in Britain were female and poor, a good number in continental Europe were male and rich, and this demographic distribution may have been not accidentally related to the more liberal laws of property seizure in continental Europe (Currie, 1968).

The medieval Church did recognize demon possession, which it distinguished from witchcraft. However, most clerics were skeptical of purported cases of demon possession, and there were few cases of prosecution because they believed possession to be a rare occurrence. Exorcisms were also infrequent and usually only performed in cases of convulsion and incoherence of speech (Neugebauer, 1978). Saint Francis of Assisi (c. 1182–1226) employed a variety of tests to discriminate the possessed from the psychologically disturbed, based upon the attributed powers and responses of demons (Kemp, 1990). Philipp Melanchthon (1497–1560) tried to achieve the same goal by splashing suspected persons with vials of ordinary well water and holy water. He judged only those few whose violent response was restricted to holy water to be possessed (Bodin, 1975).

The officers of the Inquisition were usually friars of the Franciscan order, founded by Saint Francis in the 13th century. The Franciscan order, like the Dominican order, was created in response to the Catharian heresy, which many believed could be countered only by clerics who preached orthodoxy while living lives of poverty and austerity. One of the paradoxes of the medieval period was that, while the Franciscans staffed the most oppressive office of the Church, their order also produced open-minded theorists such as Roger Bacon (c. 1214–1292), Duns Scotus (c. 1265–1308), and William of Occam (c. 1280–1349), whose work anticipated the scientific revolution of the 16th and 17th centuries (Kemp, 1990).

Natural Fools and Accidie It is commonly supposed that the medievals had little understanding of psychological disorders and consequently treated those suffering from them in cruel and barbaric ways. It is often righteously assumed that it was only in the 20th century that scientific psychology developed an adequate classification system for psychological disorders and developed effective and humane means of treating them.

Yet medieval theories of psychological disorders were quite various. Most disorders were attributed to constitutional or environmental brain damage, as in the case of those identified as “natural fools” and those whose disorders were attributed to accidents such as blows to the head (Spanos, 1978). Psychological disorders were also attributed to the imbalance of humors brought about by noxious substances such as strong wine; to emotional stress induced by overzealous work or study; and to various psychological and social causes, such as

marriage problems, frustrated love or fortune, failed ambition, guilt, jealousy, fear, bereavement, economic problems, discord between parents and children, social abuse, and stigma (Neugebauer, 1978).

The medieval view of psychological disorders was neither narrow nor conceptually unsophisticated. To take but one example, in the fifth century, the Christian theologian John Cassian listed **accidie** as the eighth deadly sin. This was a form of depression characterized as a debilitating form of apathy or disgust with life, which came to be known as the “noonday sickness,” because it drew monks away from their midday prayers. This form of depression was distinguished from the form of depression involving sadness associated with personal loss or feelings of inadequacy, which came to be known as melancholie. These different forms of depression, which were distinguished by Aristotle and Saint Paul, who thought only the former sinful, were much discussed in medieval times (Altschule, 1965). As late as the 17th century, the English physician Richard Napier (1559–1634) distinguished cases of accidie from melancholie and noted that the leisured upper classes suffered from it more frequently than the laboring lower classes (MacDonald, 1981).

Eventually the term and diagnosis fell out of use, and it was dropped from the list of deadly sins. Few people complain of accidie these days, and it is not recognized in any edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM). Nonetheless, the medievals may not have been wildly off the mark. The psychiatrist Robert Findley-Jones (1986) has suggested that the General Health Questionnaire and the Present State Examination can be employed to discriminate accidie from regular depression and that accidie appears to be especially prevalent among housewives and the unemployed (in Melbourne, Australia, at least).

Medieval modes of treatment tended to be eclectic, ranging from rest and relaxation, controlled diet, music, medicines, and folk-compounds to bloodletting, purgatives, amulets, counseling, and prayer. Most of these were based upon a holistic conception of health derived from Hippocrates. It was only in the 18th and 19th centuries that “scientific” treatments such as spinning, water-dousing, and electrical stimulation became popular, along with primitive and often dangerous experiments in psychopharmacology, involving iron, arsenic, and strychnine (Jackson, 1986). The medievals generally treated those suffering from psychological disorders with the level of sympathy and care appropriate to these harsh times.

Empiriks

Although medieval Christian scholars were not actively hostile to science, they did little to promote it. Throughout the medieval period, real opportunities for the advancement of science were frequently not exploited, not because of clerical interference, but out of a general lack of interest in pursuing theoretical questions empirically.

A good illustration of this was the failure to empirically evaluate the theories of Galen, whose anatomy and physiology were taught at most medical schools

during the medieval period. To take but one example, Galen had described a system of blood vessels at the base of the brain in humans and animals known as the *rete mirabile* (the marvelous net), which were believed to refine the animal spirits in the brain:

The plexus called retiform [rete mirabile] by anatomists is the most wonderful of the bodies located in this region. It encircles the gland [the hypophysis] itself and extends far to the rear; for nearly the whole base of the encephalon has this plexus lying beneath it.

—(*On the Usefulness of the Parts of the Body*, 1, p. 430)

His description of this system was regularly repeated in medieval medical texts (Kemp, 1990). However, fairly elementary neurophysiological examination would have established that this system does not exist in humans, although it does in ungulates, such as sheep and goats, upon which Galen practiced his dissections. Vesalius demonstrated this in the 16th century, when Padua and other Italian medical schools led the scientific revolution in medicine. Although dissection was not systematically prohibited or suppressed, and indeed was required in many medical schools, few physicians in medieval times seem to have bothered to empirically check the adequacy of Galen's account, just as few natural philosophers before Galileo seem to have bothered to empirically check the adequacy of Aristotle's false but intuitively plausible theory that bodies of different weight fall with different velocities.

For the medievals, theoretical knowledge was based upon scholarly tradition. Medical degrees at Oxford, for example, were awarded on the basis of three public lectures on the works of Galen (Kemp, 1990), not upon hours spent in biology labs or anatomy classes. If you were injured and needed surgical treatment during the medieval period, you did not consult a physician. You went to a butcher, one skilled in the practical art of the knife, or, as they were called in those days, an **empirik**. They were originally treated as charlatans, who ignored scientific theory and based their practice on observation alone.

Science as we know it today came about when theorists also became empiriks, when they began to subject their own and their predecessors' theories to empirical tests. This is what distinguishes the practice of the pioneers of the scientific revolution in Europe in the 16th century, such as Galileo and Vesalius, and eventually the practice of the first truly scientific psychologists.

Anticipations

The general tenor of the medieval mind was nonempirical: Theories were held to derive their support from tradition and scripture. Still, there were a number of medieval theorists who advanced principles of scientific methodology that were

clearly ahead of their time and which have a distinctly modern resonance. Robert Grosseteste (c. 1168–1253), chancellor of the University of Oxford, produced detailed commentaries and analyses of Aristotle's *Posterior Analytics* and *Physics* and wrote extensively on the logic of the confirmation and falsification of scientific theories.

Roger Bacon (c. 1214–1292), a Franciscan who studied at Oxford and Paris (where he later taught), also wrote extensive commentaries on Aristotle and strove to achieve a unification of the various sciences of his day. In his *Opus Magnus* Bacon put forward what he called the first and second **prerogatives of experimental science**. He maintained that any theory developed to accommodate a range of observations should be further tested via additional novel predictions derived from the theory and that experimentation, in the form of controlled intervention, can augment the naturalistic observational basis of scientific theories. Bacon is often unfairly characterized as a necromancer (an enchanter) because of his interest in alchemy and his extravagant claims about its achievements. However, he clearly recognized the need for theoretical science to unite with technical craft traditions such as alchemy, which represented one of the few approximations to empirical science in medieval times.

Duns Scotus (c. 1265–1308), a Franciscan who studied at Oxford and Paris, and William of Occam (c. 1280–1349), a Franciscan who studied at Oxford, described the methods of comparative causal analysis later known as “Mill’s methods.” William of Occam enunciated a general principle of theoretical economy that has come to be known as **Occam’s razor**: “Entities are not to be multiplied beyond necessity.” Although the principle was originally introduced within the medieval debate about the ontological status of universals, it applies generally to the evaluation of any set of competing theories.

Occam claimed that one ought not to postulate any more entities or degrees of complexity than are necessary to explain a range of phenomena in any domain. When two or more theories are equivalent in terms of the empirical data they predict and purport to explain, Occam’s razor reasonably enjoins us to choose the simplest theory. This principle eventually found its psychological expression in the methodological prescription formulated by the English comparative psychologist Conwy Lloyd Morgan (1852–1936), which came to be known as **Morgan’s canon**. Morgan claimed that psychologists should not explain animal behavior by reference to complex cognitive states if it can be explained in simpler terms, such as in terms of instincts or learned habits (Morgan, 1894/1977).

These theorists were ahead of their time, and their work had little immediate impact. Few rushed to implement their principles, and many of them failed to follow their own methodological prescriptions, often regressing to appeals to tradition or the “naturalness” of their theories. Most continued to assimilate Aristotelian philosophy to Christian theology, and their development of the more empirically oriented elements of Aristotle’s scientific writings brought them into

conflict with the Church authorities. Bacon was confined for a number of years, and Occam fled to Bavaria when the Inquisition examined his writings, although he escaped condemnation.

THE END OF THE MEDIEVAL PERIOD

The end of the medieval period is conventionally dated by the fall of Constantinople to the Turks in 1453. This date is somewhat arbitrary. The forces of change were in motion long before, and recognizable anticipations of the scientific revolution can be traced back as early as the 12th century. Yet the date is not inappropriate. By 1453 the world was changing fast and expanding rapidly. Around this date Johann Gutenberg (c. 1397–1468) printed his first Bible. Forty years later Christopher Columbus (c. 1451–1506) discovered the Americas. The Renaissance and Reformation were in full swing, and the scientific revolution was about to begin.

DISCUSSION QUESTIONS

1. The theories of early Greek scientists had few practical or technological applications. The Romans were very practical and technological, but seemed constitutionally uninterested in science. Is there no connection between science and technology? In considering this question, remember that many 20th-century theorists saw scientific psychology as the basis of a technology of social control or “social engineering,” through forms of education and treatment.
2. Theorists such as Plotinus and Avicenna followed Plato in maintaining that the psyche is immaterial and that some forms of knowledge (such as knowledge of mathematics and moral principles) are innate. Theorists such as Averroës and Aquinas followed Aristotle in maintaining that the psyche is materially incarnated and that there is no innate knowledge. Is there any connection between their views on the psyche and their views on the possibility of innate knowledge?
3. Have you ever experienced a debilitating emotion akin to *accidie*? Does the medieval characterization of this emotion bear any resemblance to contemporary clinical phenomena? Does the fact that we no longer talk about *accidie* mean that we no longer experience it?
4. Think of the scientific theories you know, in psychology and other natural and social sciences. Are they formulated, or formulable, as sets of axioms and theorems, in the manner in which they were presented by early theorists

such as Euclid and Archimedes and by later natural scientists and psychologists such as Isaac Newton and Clark L. Hull? Does such axiomization represent a desirable ideal for scientific theories (promoting scientific rigor) or an overly formal constriction (impeding the fertile development of theories)?

5. Consider the fact that the medieval theory of the “inner senses” was advocated by both materialists such as Averroës and Aquinas and by substance dualists such as Augustine and Avicenna. In this instance, commitment to an immaterial soul seems to have had no significant implications for specific psychological theories of perception and cognition. Does it have any implications for psychological theories?

GLOSSARY

accidie A form of depression recognized by ancient and medieval theorists, which they characterized as a debilitating form of apathy or disgust with life.

Averroës heresy Averroës’s claim that the active reason in all humans is numerically identical. The Christian Church condemned it as heresy in 1270.

Catharian heresy The belief that since Christ was poor the Church should abandon its wealth.

choleric type According to Galen, the irritable and emotional personality type with an excess of yellow bile.

Cynics The followers of Antisthenes and Diogenes, who rejected classical learning and conventional morality and advocated a primitive and independent lifestyle.

empirik The medieval term for a butcher, skilled in the practical art of surgery.

Epicureanism Philosophy of happiness based upon moderation developed by Epicurus and his Roman disciple Lucretius.

flying man argument Avicenna’s argument in support of substance dualism.

inner senses A medieval psychological theory that was an amalgamation of the psychology of Aristotle and the neurophysiology of Galen. The “inner senses” were identified as perceptual and cognitive faculties located in the ventricles of the brain.

Inquisition Office created by Pope Gregory IX in 1233 to combat heresy; its officers were generally friars of the Franciscan order.

intentionality The directedness or “aboutness” of psychological states such as thoughts, emotions, motives, and memories.

medieval period The period from approximately 500 to 1600 CE.

melancholic type According to Galen, the sad type of personality with an excess of black bile.

Morgan's canon Methodological prescription, advanced by the comparative psychologist C. Lloyd Morgan, that psychologists should not explain animal behavior by reference to complex cognitive states if it can be explained in simpler terms.

Neoplatonism Theories developed in the early years of the Roman Empire that emphasized the mystical and spiritual elements of Plato's philosophy.

Occam's razor Principle advanced by William of Occam, according to which no more entities or degrees of complexity should be introduced in a theoretical explanation than are necessary to explain the range of data in any domain. When empirically equivalent theories compete, Occam's razor enjoins us to choose the simplest theory.

phlegmatic type According to Galen, the slothful personality type with an excess of phlegm.

prerogatives of experimental science Methodological principles advanced by Roger Bacon, who maintained that theories should be evaluated by reference to their novel predictions and that experimentation should augment naturalistic observation.

sanguine type According to Galen, the cheerful personality type with an excess of blood.

scholasticism The term used to describe medieval attempts to integrate Aristotelian philosophy and Christian theology.

Skeptics The followers of Pyrrho, who repudiated all pretensions to knowledge.

Stoicism The philosophy of life advocated by Zeno of Citium, in which the good life is identified with acceptance of one's fate in a determined world.

substance dualism Theory that the soul (or mind) is a special and simple spiritual substance distinct from material substance, which can survive bodily death.

ventricles The fluid-filled cavities of the brain identified by Galen, which were held to be the location of perceptual and cognitive faculties according to the medieval theory of the "inner senses."

REFERENCES

- Abel, D. C. (1995). Intellectual substance as form of the body in Aquinas. *American Catholic Philosophical Quarterly*, 69 (Suppl.), 227–236.
- Alexander, F. G., & Selesnick, S. T. (1966). *The history of psychiatry: An evaluation of psychiatric thought and practice from prehistoric times to the present*. New York: Harper & Row.
- Altrocchi, J. (1980). *Abnormal behavior*. New York: Harcourt Brace Jovanovich.

- Altschule, M. D. (1965). Acedia [accidie]: Its evolution from deadly sin to psychiatric syndrome. *British Journal of Psychiatry*, 111, 117–119.
- Augustine, Saint. (1958). *On Christian doctrine*. (D. W. Robertson, Jr., Trans.) Indianapolis: Bobbs-Merrill. (Original work from fourth and fifth centuries CE).
- Augustine, Saint. (1972). *The city of God* (H. Bettenson, Trans.). New York: Penguin. (Original work from fifth century CE)
- Bodin, J. (1975). *Colloquium of the seven about secrets of the sublime* (Marion Kuntz, Trans.). Princeton: Princeton University Press.
- Brentano, Franz. (1995). *Psychology from an empirical standpoint*. London: Routledge. (Original work published 1874)
- Bullough, V. L. (1958). Medieval Bologna and the development of medical education. *Bulletin of the History of Medicine*, 32, 201–215.
- Cohen, N. (1975). *Europe's inner demons*. London: Chatto/Heinemann.
- Coulton, G. G. (1961). *Medieval panorama: Vol II. The horizon of thought*. London: Collins.
- Currie, E. P. (1968). Crimes without criminals—Witchcraft and its control in Renaissance Europe. *Law and Society Review*, 3, 1026.
- Davies, B. (1992). *The thought of Thomas Aquinas*. Oxford: Oxford University Press.
- Demaitre, L. (1975). Theory and practice of medical education at the University of Montpellier in the thirteenth and fourteenth centuries. *Journal of the History of Medicine and Allied Sciences*, 30, 103–123.
- Findley-Jones, R. (1986). Accidie and melancholie in a clinical context. In R. Harré (Ed.), *The social construction of emotion*. Oxford: Blackwell.
- Galen. (1968). *On the usefulness of the parts of the body* (Vols. 1–2; M. T. May, Trans.). Ithaca, NY: Cornell University Press. (Original work from second century CE)
- Goodman, L. E. (1992). *Avicenna*. New York: Routledge.
- Grane, L. (1970). *Peter Abelard's philosophy and Christianity in the Middle Ages* (F. Crowley & C. Crowley, Trans.). New York: Harcourt, Brace & World.
- Green, C. D. (2003). Where did the ventricular localization of mental faculties come from? *Journal of the History of the Behavioral Sciences*, 39, 131–142.
- Gutas, D. (1988). *Avicenna and the Aristotelian tradition*. Leiden: Brill.
- Haldane, J. (1992). Aquinas and the active intellect. *Philosophy*, 67, 199–210.
- Harvey, E. R. (1975). *Inward wits*. London: Warburg Institute, University of London.
- Hobbes, T. (1966). *On human nature*. In W. Molesworth (Ed.), *The English works of Thomas Hobbes*. Darmstadt, Germany: Scientia Verlag Aalen. (Original work published 1640)
- Hull, C. L. (1943). *Principles of behavior*. New York: Appleton-Century-Crofts.
- Jackson, S. W. (1969). Galen—On mental disorders. *Journal of the History of the Behavioral Sciences*, 5, 365–384.

- Jackson, S. W. (1986). *Melancholia and depression: From Hippocratic to modern times*. New Haven, CT: Yale University Press.
- Kemp, S. (1990). *Medieval psychology*. New York: Greenwood Press.
- Kemp, S. (1997). The inner senses: A medieval theory of cognitive functioning in the ventricles of the brain. In W. G. Bringmann, H. E. Lück, R. Miller & C. E. Early (Eds.), *A pictorial history of psychology*. Chicago: Quintessence.
- Kirsch, I. (1978). Demonology and the rise of science: An example of the misperception of historical data. *Journal of the History of the Behavioral Sciences*, 14, 149–157.
- Kogan, B. (1985). *Averroës and the metaphysics of creation*. Albany: State University of New York Press.
- Leaman, O. (1988) *Averroës and his philosophy*. (2nd ed.) Oxford: Clarendon Press.
- Lindberg, D. (1968). Alhazan's theory of vision and its reception in the west. *Isis*, 58, 321–341.
- Lovejoy, A. O. (1936). *The great chain of being*. Harvard: Harvard University Press.
- MacDonald, M. (1981). *Mystical Bedlam—Madness, anxiety and healing in seventeenth-century England*. Cambridge: Cambridge University Press.
- Maher, W. B., & Maher, B. A. (1985). Psychopathology: 1. From ancient times to the eighteenth century. In G. A. Kimble & K. Schlesinger (Eds.), *Topics in the history of psychology* (Vol. 2). Hillsdale, NJ: Erlbaum.
- Morgan, C. L. (1977). *Introduction to comparative psychology*. In D. N. Robinson (Ed.), *Significant contributions to the history of psychology, 1750–1920*. Series D: *Comparative psychology* (Vol. 2). Washington, DC: University Publications of America. (Original work published 1894)
- Neugebauer, R. (1978). Treatment of the mentally ill in medieval and early modern England: A reappraisal. *Journal of the History of the Behavioral Sciences*, 14, 158–169.
- Newton, I. (1969). *Mathematical principles of natural philosophy*. (F. Cajori, Trans.). New York: Greenwood Press. (Original work published 1687)
- O'Meara, D. J. (1995). *Plotinus: Introduction to the Enneads*. Oxford: Oxford University Press.
- Pyenson, L., & Sheets-Pyenson, S. (1999). *Servants of nature: A history of scientific institutions, enterprises, and sensibilities*. New York: Norton.
- Rahman, F. (Trans.) (1952). *Avicenna's psychology*. London: Oxford University Press.
- Rahman, F. (Trans.). (1958). *Shifa De Anima (Avicenna)*. Oxford: Oxford University Press.
- Roach, M. (2003). *Stiff: The curious lives of human cadavers*. New York: Norton.
- Robertson, D. W., Jr. (1972). *Abelard and Heloise*. New York: Dial Press.
- Schoeneman, T. J. (1977). The role of mental illness in the European witch hunts of the sixteenth and seventeenth centuries: An assessment. *Journal of the History of the Behavioral Sciences*, 13, 337–351.

- Scott, R. (1971). *Discovery of witchcraft*. New York: Walter J. Johnson. (Original work published 1584)
- Smith, A. M. (1996). Ptolemy's theory of visual perception. *Transactions of the American Philosophical Society*, 86, Pt. 2.
- Spanos, N. P. (1978). Witchcraft in histories of psychology: A critical analysis and an alternative conception. *Psychological Bulletin*, 85, 417–439.
- Sprenger, J., & Kramer, H. (1989). *Malleus maleficarum* [*The witches hammer*] (M. Summers, Trans.). Mineola, NY: Dover. (Original work published 1487)
- Suinn, R. (1975). *Fundamentals of behavior pathology* (2nd ed.). New York: Wiley.
- Thorndike, L. (1944). *University records and life in the Middle Ages*. New York: Cambridge University Press.
- Trevor-Roper, H. R. (1967). *The European witch-craze of the 16th and 17th centuries*. Harmondsworth, England: Penguin.
- von Straden, H. (1989). *Herophilus: The art of medicine in early Alexandria*. Cambridge: Cambridge University Press.
- Wedel, T. (1968). *The medieval attitude towards astrology, particularly in England*. Hamden, CT: Archon.
- Weyer, J. (1563). *Deception of demons*. London.
- Zax, M., & Cohen, E. L. (1976). *Abnormal psychology*. (2nd ed.). New York: Holt, Rinehart & Winston.



The Scientific Revolution

ON OCTOBER 11, 1572, THE DANISH ASTRONOMER TYCHO BRAHE (1546–1601) observed a bright new object in the evening sky. This was “a miracle indeed,” since this object did not move against the background of fixed stars and must itself have been a star. Yet Aristotle had taught that everything in the celestial region, the sphere of fixed stars, was perfect and unchanging. Brahe’s observation of what we now believe to have been a supernova (a new star) was one of the many developments that led to the eventual overthrow of the Aristotelian geocentric (Earth-centered) astronomical system and the medieval worldview based upon it.

By the 14th century, the social, political, and intellectual order of the medieval world had begun to break down. Increased urbanization and the return to a money economy eroded the structure of the feudal system, and the rise of nation-states undermined the political authority of the papacy. Intermittent wars between the emerging nation-states led to a severe economic depression. This was followed by the plague of 1348–1350, later known as the “Black Death,” which decimated the European population and bred doubt and resentment against the medieval Church, the dominant authority. Although the Church embraced Aristotle’s philosophy, the threat posed by its naturalism and rationalism generated dissent and division, leading initially to attempts to divorce the separate realms of faith and reason and then to the autonomous emergence of naturalistic empirical science.

Various developments contributed to the transformation of the intellectual landscape. Marco Polo’s (1254–1324) exploration of China, Christopher Columbus’s (1451–1506) discovery of America in 1492, and Magellan’s (1480–1521) circumnavigation of the globe expanded the horizons of the known world. Perhaps the most significant development was the invention of printing and the consequent transformation of communication. In the city of Mainz in southern Germany, Johann Gutenberg (c. 1397–1468) created movable type and published an edition of the Bible in 1450. The consequent explosion in printed works expanded intellectual horizons by broadening access to the Bible and classical works. By 1500, about 8 million volumes had been printed (Pyenson & Sheets-Pyenson, 1999); by 1600, about 20 million, with over a dozen presses established in European cities (Foote, 1991). The critical interpretation of these works by